HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 08 Name: _____ Marks: / 35

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Factorize

   (a) $6r^2 - 13rs - 28s^2$,

   (b) $4r - 14s + 6r^2 - 13rs - 28s^2$.

(3 marks) (DSE-2024 PI Q03)

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2. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100 ? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

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3. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

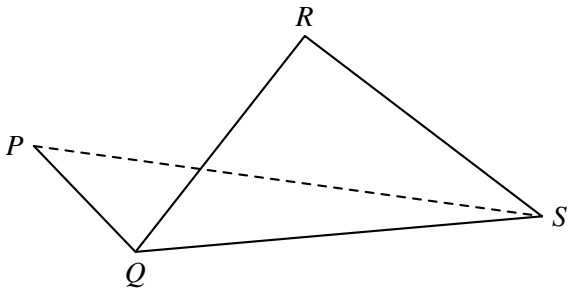


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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4. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

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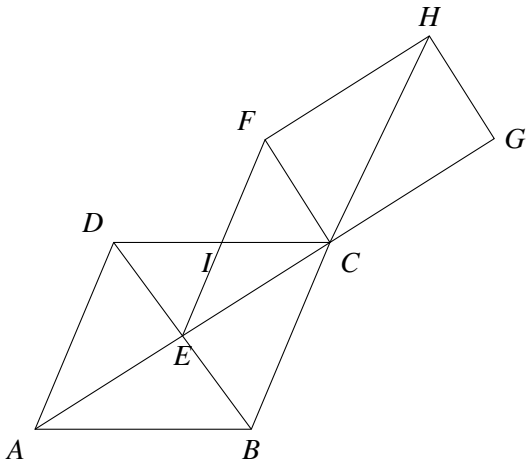
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5. In the figure, $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Let F be a point such that $BH//EF$ and $CFHG$ is a rectangle, where G and H are points lying on AC produced and BC produced respectively. Denote the point of intersection of CD and EF by I. Which of the following must be true?

I. $CI = FI$

II. $\angle ABE = \angle GCH$

III. $\triangle ADE \cong \triangle HCF$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q21)

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6. Denote the circle $3x^2 + 3y^2 - 6x + 12y - 4 = 0$ by C. Which of the following are true?

I. The origin lies inside C.

II. The circumference of C is less than 16.

III. The perpendicular distance from the centre of C to the x-axis is 2.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q27)

I can finish _____ questions

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